SUKKUR IBA UNIVERSITY
DEPARTMENT OF COMPUTER SYSTEMS ENGINEERING

FYDP Progress Report

Project Title:
SafeDrive AI: Intelligent dashboard device for driver anomaly detection

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Progress Report Submission Date:

Details of FYDP Work:

Final Year Design Project Work	Hardware Based (%)	Software Based (%)
	60%	40%

	In term of (%)	Description
Executed Work	**90%**	• Applied foundational data augmentation techniques to increase dataset variability and effectively address class imbalance. • Trained the YOLOv8n-cls model using multiple hyper parameter configurations, including variations in batch size, learning rate, and training epochs, to optimize classification performance. • Observed that the YOLOv8-cls model delivered reliable and higher-accuracy results based on the conducted experiments. • Completed the full environment setup on the Raspberry Pi for on-device inference. • Deployed and tested the trained YOLOv8-cls model on the Raspberry Pi using a sample video to validate operational performance. • Used Gamma Correction to darken mid-tones and crush shadows creating a high-contrast looks of low-light environments. • Also use channel-specific scaling and desaturation to mimic conditions where blue tones dominate as the human eye loses color sensitivity. • Retrained the YOLOv8-cls model using the expanded dataset and validate the updated performance metrics. • Converted the retrained model to NCNN format to enable deployment on the Raspberry Pi. • Tested the model in real-time environment in car and observed the behavior of the system. • Physically mount and connect the Camera Module to the Raspberry Pi for live video input. • Wire the physical speaker to the Raspberry Pi to enable real-world alerts. • Conduct final "in-vehicle" testing to calibrate the camera under real sunlight and night-driving conditions.
Remaining Work	**10%**	• Enclose the Entire System and give it a physical shape of a working device. • Configure the entire software stack as a Linux System

Comments by Supervisor/Co-Supervisor:

Is the FYDP Project/Thesis ready for Evaluation I/II/III/final presentation? No ☐ Yes ☐

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Signature

Supervisor

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